Domain Adaptation of Pretrained Convolutional Neural Networks via Nonlinearity Calibration

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Abstract

Pretrained convolutional neural networks (CNNs) are widely used for transfer learning, but full fine-tuning is impractical in the medical imaging domain due to data scarcity. Parameter-efficient fine-tuning (PEFT) methods are lightweight alternatives that update only a small subset of model parameters to reduce computational load and the volume of training data. In this paper, we investigate *learnable activation functions* as a novel PEFT strategy for adapting readily available pretrained CNNs to the medical domain. In contrast with traditional PEFT approaches, we hypothesize that we can replace the static activation function of a pretrained model with a dynamic function with learnable parameters, and then learn those parameters on a small medical dataset. To show the effectiveness of the proposed method, we extend a ResNet-50 architecture with three activation variants: Parametric ReLU (PReLU), SwishLearn, and a novel highly non-linear activation (KGLAP), and benchmark them against baseline methods trained on a brain tumour MRI classification task. The results show that learnable activations are a viable PEFT methodology, as they deliver F1 and accuracy scores that match or exceed traditional approaches. Moreover, learnable activations consistently deliver lower expected calibration error than benchmark methods, positioning them as an efficient and resilient adaptation mechanism for medical imaging, where reliable confidence estimation is as critical as accuracy. Future work is required to extend this methodology to additional architectures, explore more intricate activation configurations, and develop initialization strategies.

Keywords:
CNN's, PEFT, Transfer Learning, Medical Imaging, Activation Functions

1. Introduction

Convolutional neural networks (CNNs) have become the backbone of modern image classification systems due to exceptional performance across a wide range of domains [1]. While a wealth of powerful pre-trained models are readily available for practitioners, their performance degrades when the input data diverges from the distribution they were originally trained on. This reliance limits the transferability to new input modalities, whether shifting from natural to domain-specific images, or more broadly, from visual to non-visual domains [2, 3] (See Figure 1).

A central challenge in deep learning is to extend the adaptability of pretrained CNNs beyond their original domain. This includes specialized visual tasks such as medical or remote sensing imagery [3], and ultimately to non-visual modalities like time series or spectrograms [4].

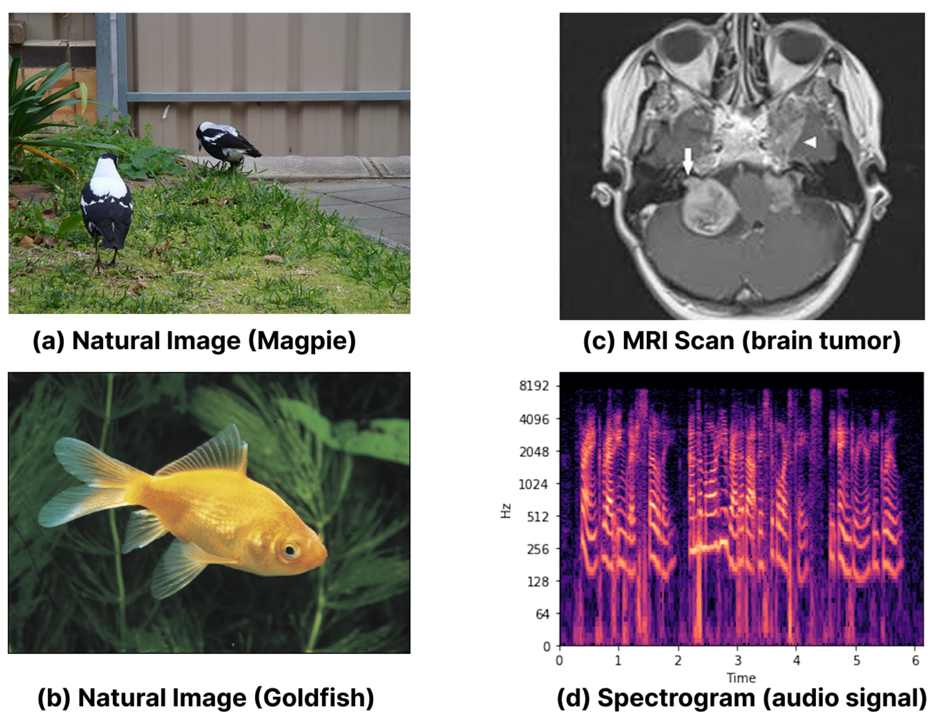

Figure 1: Illustration of the domain gap between pretraining and downstream applications. (a–b) Natural images from ImageNet used during pretraining. (c) A brain MRI image used in medical imaging tasks. (d) A spectrogram representing a non-visual modality.

In the field of medical imaging, CNNs have shown the potential to match or exceed expert-level diagnostic accuracy. Esteva et al. [5] demonstrated that a CNN could classify skin cancer at a level comparable to board-certified dermatologists. Their approach began with an Inception v3 model pretrained on ImageNet, a large-scale dataset of 1.2 million natural images spanning 1,000 categories. They then fine-tuned the entire network on a dataset of approximately 129,450 clinical images covering over 2,000 skin conditions, achieving area under the ROC curve (AUC) scores of 0.94 for melanoma and 0.96 for carcinoma. Although the expert-level performance accomplished by Esteva et al. [5] demonstrated the potential of CNNs to assist diagnosticians, several key challenges remain for the widespread adoption of such models in medicine. The first roadblock researchers face is the limited availability of suitable medical data. Creating large and high-quality datasets often requires expert annotations by radiologists, pathologists, or dermatologists, and may involve pixel-level detail for tasks like segmentation, making it expensive, slow, and labour-intensive [6]. Furthermore, privacy regulations limit data sharing between institutions, which prevents the aggregation of large, diverse datasets and hinders reproducibility [7]. Moreover, medical AI models tend to be highly domain-specific. Differences in imaging protocols, scanners, patient demographics, and institutional practices can cause significant distribution shifts, making it difficult for a model trained in one setting to generalize to another [8, 9]. Finally, replicating large-scale efforts like Esteva et al.'s requires substantial resources, including multidisciplinary expertise, institutional partnerships, and high-performance computing infrastructure, making such efforts prohibitively costly for most researchers or clinical teams [10].

Together, these challenges constrain the data and computational resources available to clinical teams, limiting the viability of traditional full fine-tuning strategies such as those employed by Esteva et al. A popular solution is to use parameter-efficient fine-tuning (PEFT) methods to calibrate pretrained models using minimal training resources. Unlike traditional full fine-tuning, which updates a large portion of model parameters, PEFT methods achieve adaptation by introducing a small number of trainable parameters while keeping the pretrained backbone largely frozen [11, 12]. These strategies enable effective fine-tuning even on smaller datasets, as they have been shown to mitigate overfitting and generalize better than full fine-tuning under data-scarce conditions [12, 13].

Our research explores a novel approach to PEFT by introducing learnable activation functions into the architecture of pretrained CNNs. Unlike previous PEFT methods that insert structural modules or low-rank projections, our strategy modifies the nonlinearities in the network layers to make them trainable. Learnable activation functions have been shown to improve neural network expressivity and performance in supervised learning tasks, outperforming traditional fixed activations across a range of architectures and datasets [14, 15]. Our research is among the first to investigate learnable activations as a mechanism for parameter-efficient adaptation in transfer learning. By replacing static activation layers with lightweight, trainable alternatives, we aim to create a flexible and computationally efficient calibration mechanism that enables robust adaptation to new data distributions under constrained training regimes. In this work, we extend a pretrained ResNet-50 backbone [16, 17] with several simple adaptive activation functions, as well as a novel highly non-linear activation function inspired by the Klein–Gordon dynamics explored by Yin et al. [18] in the context of physics-informed neural networks (PINNs). We benchmark our approach against modern PEFT baselines, including LoRA-C [19] and Conv-Adapter [20] on a low-resource medical image classification task.

2. Literature Review

Our work builds upon several research streams to significantly reduce the data and computational requirements for training deep learning models for data-scarce, high-impact tasks, such as medical image analysis. While traditional transfer learning strategies leverage pretrained models by fine-tuning portions of the network on domain-specific data, many still require substantial labelled datasets and training budgets. To address this, we explore a novel PEFT strategy that tunes the nonlinearities of the model through learnable activation functions. By integrating these lightweight trainable components into a frozen backbone, we enable robust adaptation to new data distributions while preserving efficiency. We benchmark our approach against established PEFT methods to assess both discriminative ability and confidence calibration.

2.1. Transfer Learning, Calibration, and Robustness in CNNs

2.1.1. Domain Limitations of CNNs

Despite the impressive performance of modern CNNs on benchmark datasets, deploying them in real-world applications remains costly. Training a model from scratch requires massive resources. Research has demonstrated that state-of-the-art models often require hundreds of millions of labelled images [21, 22]. Acquiring such data is expensive, particularly in fields such as medical imaging, where annotations must be created by trained experts [23]. Computational cost presents another major obstacle, as it has even been projected that if current trends continue, the cost of training frontier AI models could exceed the total economic output of entire nations [24]. Even moderately sized models regularly take weeks or months to train on large-scale GPU infrastructure [21], limiting prototyping and development speeds. Moreover, building and deploying deep learning systems requires specialized expertise in machine learning, infrastructure engineering, and regulatory compliance, making these efforts accessible only to well-resourced institutions [25]. The compute and data-intensive approach to scaling model performance is increasingly economically unsustainable for most researchers, particularly those outside the deep learning community, who aim to apply these models in domain-specific contexts.

These costs are especially prohibitive in many high-impact domains such as medical imaging, where data scarcity and logistical barriers make training deep learning models from scratch impractical. The creation of high-quality labelled datasets in this field often requires annotations from domain experts such as radiologists or dermatologists, and may involve fine-grained segmentation or multi-class labelling [23]. This process is slow, expensive, and difficult to scale. Data sharing is further constrained by medical privacy regulations, which limit the ability to aggregate diverse datasets across institutions and hinder reproducibility [7]. Even when data is available, medical imaging models are highly sensitive to differences in scanner hardware, patient populations, and clinical workflows, all of which contribute to significant distribution shifts between training and deployment environments [8, 9]. As a result, models trained in one hospital or region often fail to generalize to others. Replicating large-scale efforts like those of Esteva et al. requires resources that most clinical teams lack access to including substantial institutional investment, specialized personnel, and computing infrastructure [10]. This underscores the need for more efficient and accessible approaches to model adaptation in the medical domain.

2.1.2. Transfer Learning

The most practical way to circumvent the high cost of training from scratch is to leverage models pretrained on large, generic datasets. These models encapsulate rich visual representations and are readily available thanks to large-scale institutional efforts. However, their effectiveness typically relies on a close match between the training and deployment domains. Deep learning models, including CNNs, are highly sensitive to distribution shifts and often experience significant performance degradation when applied to data that differs from what they were originally trained on [3, 2]. This presents a major obstacle to reuse, particularly in specialized domains like medical imaging or remote sensing [3]. Expanding the adaptability of pretrained models across modalities, from natural to medical images, or from images to time series or spectrograms, would unlock enormous value for downstream applications [4]. A common solution to the prohibitive costs of training from scratch is transfer learning. This method enables practitioners to fine-tune pretrained models using relatively small amounts of task-specific data. This is typically achieved by freezing the early layers of a network responsible for capturing generic low-level features such as edges and textures, and retraining only the later layers which encode higher-level semantic information relevant to the target task [26, 3]. The extent to which layers are frozen or unfrozen depends on the similarity between source and target domains, as well as on available compute and data resources. This strategy has proven effective in many domains, as it leverages the broad visual knowledge already encoded in models trained on large-scale datasets like ImageNet, while adapting only the task-specific components [27].

Despite its popularity, transfer learning remains an imperfect solution with several known challenges. The following limitations can significantly reduce transfer learning effectiveness, especially in data-scarce or high-risk domains:

- **Domain shift:** Transfer learning assumes that the source and target domains are similar. When this assumption is violated, performance can degrade sharply. Models may fail to generalize to new distributions and exhibit unjustified confidence in their predictions [8, 28]. If the domains are radically different, pretrained features may transfer poorly or not at all.

- **Overfitting:** Large models with many trainable parameters can easily overfit small target datasets, memorizing the training data and failing on unseen samples. This often leads to highly confident but incorrect predictions [12].

- **Hyperparameter sensitivity:** The success of transfer learning depends on numerous fine-tuning decisions, including learning rates, which layers to freeze, and data augmentation strategies. These require extensive experimentation and domain expertise to optimize effectively [29, 13].

These challenges often lead to poor generalization and miscalibration, where the model's predicted probabilities do not align with the actual likelihood of correctness. This is particularly concerning in high-stakes domains like medicine, where decisions informed by inaccurate confidence estimates can have serious clinical consequences. A holistic set of evaluation metrics such as precision, recall, area under the ROC (AUC), and expected calibration error (ECE) are commonly used to quantify these issues [28]. Collectively, these limitations highlight the need for more efficient alternatives that preserve the benefits of transfer learning while improving robustness and calibration under constrained training conditions.

2.2. Parameter-Efficient Transfer Learning (PEFT)

Parameter-Efficient Fine-Tuning (PEFT) provides an efficient alternative to full model fine-tuning through transfer learning by updating only a small parameter subset of a pretrained model [11, 13]. Two widely used strategies are adapter insertion, where lightweight trainable modules are added to frozen network layers [11], and low-rank adaptation, which injects trainable low-rank matrices alongside frozen weight matrices to produce task-specific behaviour [19]. These approaches reduce memory and compute requirements, shorten training time, and help prevent overfitting by restricting optimization to a controlled parameter subset. PEFT methods often train less than 1 % of a model's total parameters, in contrast to traditional fine-tuning, which sometimes requires updating the entire model.

While PEFT offers clear efficiency advantages, it also has notable limitations. Recent studies have shown that PEFT methods may converge more slowly than full fine-tuning in extremely low-data regimes, underperform on tasks requiring significant domain shifts or high expressive capacity, and introduce architectural complexity in implementation [30, 31, 32]. Furthermore, tuning hyperparameters such as adapter size or low-rank dimensions is often heuristic and task-specific, and the interpretability of these modules remains limited. These constraints highlight the need for adaptation techniques that retain the efficiency benefits of PEFT while improving flexibility, robustness, and calibration in real-world deployment.

2.3. Learnable Activation Functions

Activation functions are a foundational building block of all neural networks. Traditional convolutional neural networks (CNNs) rely on fixed nonlinear activation functions such as ReLU, sigmoid, or tanh to introduce nonlinearity into the model. Nonlinearities are essential as they enable deep networks to approximate complex functions while breaking linearity across layers and address issues like the vanishing gradient problem [33, 34]. However, fixed activations cannot adapt to the structure of the data or task, limiting the flexibility of the network. Learnable activation functions address this limitation by introducing trainable parameters, allowing the activation shape to adjust dynamically during training [34, 35]. Examples include Parametric ReLU (PReLU), which learns a slope for the negative input range [35], and Learnable Swish, which incorporates a trainable scaling factor to enhance expressivity [36]. These functions integrate seamlessly into standard training pipelines and are optimized via backpropagation using typical optimizers like Adam. Empirical results have shown that learnable activations can improve model expressivity, adaptability, and task-specific performance in supervised settings [34, 36]. Nevertheless, existing research has focused primarily on their use in standard supervised learning, and to the best of our knowledge, none have explored their integration into parameter-efficient transfer learning (PEFT) frameworks. Our work addresses this gap by investigating learnable activation functions as a lightweight PEFT mechanism, enabling efficient adaptation by tuning nonlinearity rather than altering network structure.

2.4. Research Gap and Contribution

Although parameter-efficient fine-tuning (PEFT) has substantially reduced the cost of adapting large models, most existing approaches continue to struggle in medical imaging, where datasets are exceedingly small and unbalanced. We argue that tuning the nonlinearity itself offers a more direct and elegant path to efficient transfer, as learnable activation functions have already been shown to accelerate convergence and enhance adaptability in standard supervised learning. Our work explores this direction by integrating learnable activation functions as a lightweight PEFT mechanism, enabling fine-grained, data-efficient adaptation that aligns naturally with the constraints of clinical imaging practice.

3. Methodology

3.1. ResNet-50-v1.5 (ImageNet Pretrained) as a Benchmark Backbone

We chose the ResNet-50 architecture as our CNN backbone due to its maturity, widespread adoption, and sequential bottleneck structure. These properties allow for straightforward integration of parameter-efficient tuning mechanisms. ResNet-50 was originally introduced by He et al. in 2015 [17]. It remains widely used for image classification and transfer learning tasks due to its efficiency and well-documented modular architecture. It is characterized by a stack of residual bottleneck blocks organized into four stages as per Figure 2. Each bottleneck block features identity or projection shortcuts that preserve gradient flow during backpropagation, making it suitable for training deep models without degradation.

The maturity and ubiquity of ResNet-50 also mean that well-developed parameter-efficient fine-tuning (PEFT) baselines such as LoRA and Conv-Adapters are readily available and compatible with its structure. All methods evaluated in this work begin with the same ResNet-50-v1.5 implementation and are initialized with pretrained ImageNet-1K weights from PyTorch. This ensures a standardized and fair comparison across all PEFT strategies tested in this study. An overview of the model architecture and parameter breakdown is provided in Figure 2.

3.2. Learnable Activation Functions as a Parameter-Efficient Tuning Mechanism

We propose a novel PEFT strategy based on introducing *learnable activations* into the final stages of a pretrained CNN. Fixed nonlinearities such as ReLU are computationally efficient but can restrict the network's adaptability to new data distributions without retraining large portions of the model. In contrast, parameterizing the activation functions provides a lightweight and structurally simple means of adaptation. Rather than adding modules or retraining high-level convolutional filters, we modulate the nonlinearities themselves. This preserves the inductive bias of the pretrained backbone while adding a compact set of tunable parameters.

Learnable Activation Placement: Learnable activation functions were inserted into the final convolutional stages of ResNet-50 to enable parameter-efficient adaptation at different depths of the network. Two insertion schemes

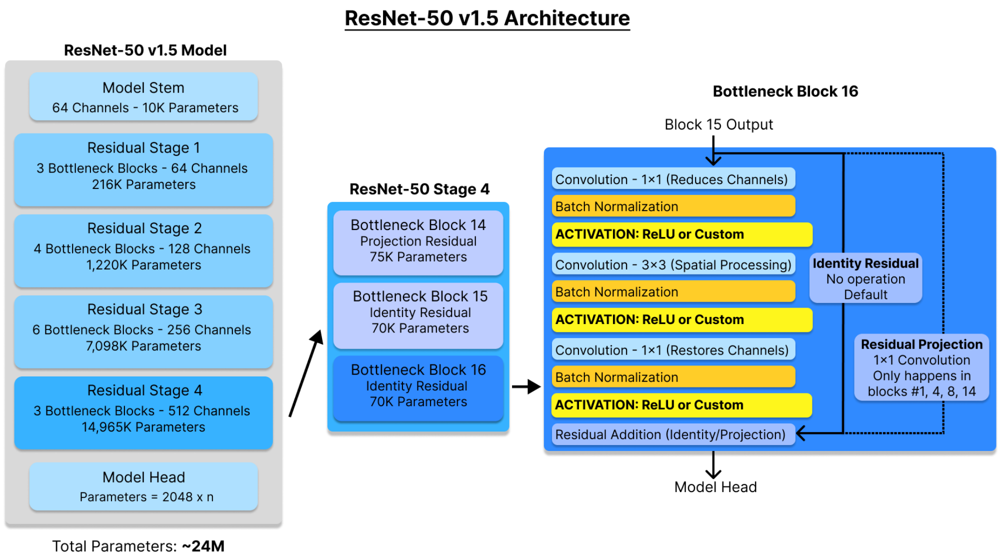

Figure 2: Overview of the ResNet-50 v1.5 architecture. The model has four main residual stages, each containing bottleneck blocks with either identity or projection shortcuts. Each bottleneck block contains a 1×1 convolution (for channel reduction), a 3×3 convolution (for spatial processing), and a 1×1 convolution (for channel restoration). Residual projection is used only at the start of each stage (Blocks 1, 4, 8, and 14), while subsequent blocks use identity shortcuts. Parameter distributions across stages are shown, highlighting that the majority of parameters are concentrated in the final residual stage. Custom learnable activations can be inserted at the ReLU sites indicated in yellow.

were investigated: (1) *Block-Shared*, in which a single set of activation parameters is shared across all channels within each residual block, and (2) *Channel-Wise*, where each feature channel learns its own activation parameters independently. Block-Shared configurations prioritize parameter efficiency and stability, while Channel-Wise variants provide greater expressive capacity at a modest increase in trainable parameters. To clearly distinguish these configurations in tables and figures, model names are reported using the format *(Parameterization Type, Stage Range)*; for example, *PReLU (Block-Shared, Stages 3–4)* or *KGLAP (Channel-Wise, Stage 4)*.

Each activation function was implemented as a differentiable PyTorch module and integrated into the autograd system. Activation parameters were trained alongside the classifier head using Adam, but grouped separately from classifier weights with distinct optimizers, enabling differential learning rates and clearer attribution of learning dynamics.

The three learnable activations evaluated are:

- **PReLU (Parametric ReLU)** [35], a simple baseline where the negative slope is a learned parameter. We initialize this slope at 0.25.

- **SwishLearn**, a scalar-parameterized variant of Swish:

$$f(x) = x \cdot \sigma(\beta x),$$

  where β is a trainable scalar initialized to 1.0. This allows the nonlinearity to interpolate between ReLU-like and smoother Swish-like behaviours.

- **KGLAP**, a physics-inspired nonlinear activation motivated by adaptive activations used in physics-informed neural networks [37]. Jagtap et al. showed that making activations trainable accelerates convergence and improves solution quality when solving nonlinear PDEs. Their adaptive activations reshape the loss landscape dynamically, helping networks capture both low- and high-frequency solution components. Inspired by this, we designed KGLAP to parallel the nonlinear Klein–Gordon PDE:

$$f(y) = y + \alpha \Delta y + \beta y + \gamma y^k,$$

  where Δy is a discrete Laplacian applied via depthwise convolution, and α, β, and γ are learnable coefficients. These terms correspond respectively to identity mapping, spatial coupling, linear response, and higher-order nonlinear interaction. This structure embeds an interpretable physics prior into the activation function, aligning with the form of governing equations known to generate complex dynamics. Parameters are initialized as $(\alpha, \beta, \gamma) = (0.1, 1.0, 0.1)$ with fixed exponent k, balancing expressivity with stability during early training.

3.3. General Training Algorithm

To ensure reproducibility and clarity in our evaluation, the complete training pipeline including initialization, optimization of activation parameters, and validation routines is formalized in Algorithm 1. This procedure applies to all learnable activation configurations evaluated in our experiments.

Algorithm 1 Training ResNet-50 with learnable activation functions

Require: training set $\mathcal{D}_{\text{train}}$, validation set $\mathcal{D}_{\text{val}}$, test set $\mathcal{D}_{\text{test}}$, class count n, patience p, max epochs T

Ensure: best model parameters $(\theta^\star, \phi^\star)$ and test metrics

procedure SETUP
  Load pretrained backbone f_θ (ResNet-50 v1.5, ImageNet-1K)
  Replace classifier head with linear layer $2048 \to n$
  Freeze all pretrained weights in θ except the new head
  Inject learnable activations at specified ReLU sites; denote their params by ϕ
  Initialize Adam optimizers $\textsc{opt}_{\text{net}}$ (head in θ) and $\textsc{opt}_{\text{act}}$ (activations ϕ)
end procedure
$L_{\text{best}} \leftarrow \infty$, $\textit{no\_improve} \leftarrow 0$
for epoch $= 1 \ldots T$ **do**
  for each $(x, y) \in \mathcal{D}_{\text{train}}$ **do**
    $\hat{y} \leftarrow f_{\theta,\phi}(x)$
    $\mathcal{L} \leftarrow \text{CrossEntropy}(\hat{y}, y)$
    $\textsc{opt}_{\text{net}}$.zero_grad(); $\textsc{opt}_{\text{act}}$.zero_grad()
    $\nabla \leftarrow \nabla_{(\theta,\phi)}\mathcal{L}$ ▷ backprop
    $\textsc{opt}_{\text{net}}$.step(); $\textsc{opt}_{\text{act}}$.step()
  end for
  $L_{\text{val}} \leftarrow \frac{1}{|\mathcal{D}_{\text{val}}|} \sum_{(x,y)\in\mathcal{D}_{\text{val}}} \text{CrossEntropy}(f_{\theta,\phi}(x), y)$
  if $L_{\text{val}} < L_{\text{best}}$ **then**
    $L_{\text{best}} \leftarrow L_{\text{val}}$; $(\theta^\star, \phi^\star) \leftarrow (\theta, \phi)$; $\textit{no\_improve} \leftarrow 0$
  else
    $\textit{no\_improve} \leftarrow \textit{no\_improve} + 1$
    if $\textit{no\_improve} \geq p$ **then**
      break ▷ early stopping
    end if
  end if
end for
Load $(\theta^\star, \phi^\star)$
Compute test metrics on $\mathcal{D}_{\text{test}}$: accuracy, AUC, ECE

3.4. Baseline PEFT Implementations

To benchmark the effectiveness of learnable activations, we compare against two established baseline PEFT methods that are structurally distinct yet architecturally compatible with ResNet-50: **Conv-Adapter** and **LoRA-C**. Both have been widely explored in the PEFT literature and insert distinct trainable structural modules into the network. The behaviour and parameter counts of these methods are highly sensitive to their respective hyperparameters; please refer to Section 4.2 for implementation details and the specific configurations used in this study.

Conv-Adapter: Introduces lightweight convolutional adapter modules into a frozen CNN [11, 20]. We adopt the `ConvSequential` variant, which follows a bottleneck design with three layers: a 1×1 projection to reduce channels, a depthwise 3×3 convolution for spatial processing, and a 1×1 expansion to restore dimensionality. Only adapter parameters are updated during training; all ResNet-50 weights remain frozen. The key hyperparameter is the *reduction ratio* γ, which determines the bottleneck width. For instance, $\gamma = 4$ reduces a 2048-dimensional input to 512 channels. Lower channel counts reduce parameter overhead but may sacrifice expressive capacity.

LoRA-C: Adapts the LoRA framework to convolutional layers by injecting trainable low-rank matrices into frozen convolutional kernels [19, 38]. For a convolutional weight tensor W, LoRA-C introduces a residual $\Delta W = AB$ with $A \in \mathbb{R}^{d\times r}$ and $B \in \mathbb{R}^{r\times k}$, where $r \ll \min(d, k)$. We apply this decomposition to all 3×3 convolutions in ResNet-50. The *rank* r controls both efficiency and expressivity: small ranks (e.g., $r = 4$ or 8) yield minimal overhead, while larger ranks increase modeling capacity at the cost of additional parameters.

ReLU FC Only: To further isolate the contribution of nonlinearity adaptation, we include a **ReLU FC Only** baseline in which all convolutional layers remain frozen and only the final fully connected (classifier) layer is trained. This setup represents the simplest and most widely used transfer learning strategy for CNNs. Because our proposed approach trains the classifier jointly with activation function parameters, this baseline serves as a lower-bound benchmark, attributing any observed performance gains in other methods specifically to the fine-tuning of intermediate activations or structural modules.

Implementation Details: All PEFT baselines were re-implemented in PyTorch according to their original specifications to ensure consistent integration and fair comparison. Only adapter, low-rank, or classifier parameters were updated during training; the rest of the ResNet-50 backbone remained frozen. All methods were trained using the same optimizer, learning rate, and early-stopping regime to isolate architectural differences. A breakdown of trainable parameter counts for each configuration is provided in Table 1.

Table 1: Trainable parameter counts for each parameter-efficient fine-tuning (PEFT) configuration evaluated. Values are reported as absolute counts and as a percentage of the total parameters in ResNet-50-v1.5 ($\sim$24.5M). This comparison highlights the lightweight nature of learnable activation functions ($< 0.06\%$ of parameters) relative to Conv-Adapters and LoRA-C, which introduce parameter counts on the order of 10^5.

Configuration	**Trainable Parameters**	**% of Total**
ReLU FC Only	8,196	0.03485%
PReLU (Block-Shared, Stages 3–4)	8,205	0.03489%
SwishL (Block-Shared, Stages 3–4)	8,205	0.03489%
KGLAP (Block-Shared, Stages 3–4)	8,223	0.03497%
PReLU (Channel-Wise, Stage 4)	9,732	0.04138%
SwishL (Channel-Wise, Stage 4)	9,732	0.04138%
KGLAP (Channel-Wise, Stage 4)	12,804	0.05444%
Conv-Adapter ($\gamma = 16$)	171,752	0.72531%
LoRA-C ($r = 4$)	445,956	1.86172%
LoRA-C ($r = 8$)	883,716	3.62301%

3.5. Evaluation Metrics

In diagnostic medical imaging, a holistic evaluation of models is paramount, as raw classification accuracy does not fully reflect diagnostic ability. Medical models must be tolerant of class imbalance, because the diagnosis of rare conditions is just as important as common ones. Additionally, correct calibration of predicted confidence scores is essential, as clinicians must be able to trust that a model's reported probabilities reflect the true likelihood of the diagnosis. To this end, we chose to evaluate the methods described above primarily using an F1 score, the area under the ROC curve (AUC), and the expected calibration error (ECE).

F1 Score: The F1 score is the harmonic mean of precision and recall, providing a balanced measure of a model's ability to correctly identify positive cases while limiting false positives. Precision is defined as $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$, the proportion of predicted positives that are correct, while recall (or sensitivity) is $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$, the proportion of true positives that are successfully detected. By combining these two quantities, the F1 score penalizes both false positives and false negatives, making it particularly suitable for diagnostic tasks where missing a true case (FN) or misclassifying a healthy patient (FP) is serious. We report the *macro-averaged* F1 score, which computes F1 for each class independently and averages them equally, ensuring that minority classes contribute as much as common ones to the overall score.

AUC: The area under the receiver operating characteristic (ROC) curve (AUC) measures a model's ability to distinguish between classes independently of any fixed decision threshold. In multiclass diagnostic classification, we compute one-vs-rest ROC curves with micro-averaging using scikit-learn's `roc_auc_score` function (`average="micro"`, `multi_class="ovr"`). A higher AUC indicates stronger discriminative ability, meaning the model more consistently ranks true disease cases above non-cases across all possible thresholds. This makes AUC a valuable indicator of general diagnostic separability, even when the optimal sensitivity–specificity trade-off varies across tumor types or clinical contexts.

ECE: Expected Calibration Error (ECE) is the most common model confidence calibration metric and quantifies the average discrepancy between predicted confidence and the real-world likelihood. It is computed by partitioning predictions into $M = 15$ equal-width confidence bins, each representing a range of model confidence values. For each bin B_m with empirical accuracy $\mathrm{acc}(B_m)$ and mean predicted confidence $\mathrm{conf}(B_m)$,

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|.$$

Lower ECE values indicate better calibration and more trustworthy probabilistic predictions—an essential property for medical applications, where overconfident errors can directly compromise patient safety and undermine clinical decision-making.

4. Dataset and Experiments

4.1. Dataset Summary

We evaluate our methodology on the publicly available *Brain Tumor MRI Dataset* from Kaggle [39], which contains 7,023 contrast-enhanced T1-weighted MRI scans. The task is framed as a **four-class classification problem**, distinguishing between **glioma (1,621, 23.1%)**, **meningioma (1,645, 23.4%)**, **pituitary tumor (1,757, 25.0%)**, and **no tumor (2,000, 28.5%)**. This represents a clinically meaningful application, as accurate tumor subtype identification is a critical step in medical diagnosis and treatment planning. All images are preprocessed with the standard ImageNet pipeline of resizing to 256×256, center-cropping to 224×224, normalization with ImageNet mean and standard deviation, ensuring compatibility with common pretrained CNN backbones. We perform stratified data splits to preserve class balance, fixing 20% of the dataset as a held-out test set and dividing the remaining 80% into training and validation subsets at an 80/20 ratio. To study data efficiency, we additionally subsample the training and validation portion at fractions of 0.005, 0.01, 0.1, 0.5, and 1.0, with distributions summarized in Table 2. While modest in scale, this dataset is representative of the quantity of annotated imaging data that might realistically be available from a single clinic or research group. This setup enables us to evaluate model performance under both extremely low-data and full-data conditions, highlighting the effectiveness of parameter-efficient transfer learning methods across different data regimes.

Table 2: Dataset splits by subset fraction and class distribution for the *Brain Tumor MRI Dataset* [39] The full dataset contains 7023 images: Glioma (G) = 1621 (23.1%), Meningioma (M) = 1645 (23.4%), No Tumor (N) = 2000 (28.5%), Pituitary (P) = 1757 (25.0%). Each subset is stratified to preserve this original class balance.

	Train					**Val**				
Subset	**Total**	G	M	N	P	**Total**	G	M	N	P
0.005	**22**	5	6	6	5	**6**	1	1	2	2
0.01	**44**	10	10	13	11	**12**	3	3	3	3
0.1	**448**	104	104	128	112	**113**	26	27	32	28
0.5	**2247**	519	526	640	562	**562**	129	132	160	141
1.0	**4495**	1037	1053	1280	1125	**1124**	260	263	320	281

Test set (constant): 1404 total [G = 324, M = 329, N = 400, P = 351].

4.2. Experimental Design

We evaluated three parameter-efficient fine-tuning (PEFT) strategies: learnable activations (our method), Conv-Adapter, and LoRA-C. All methods share the same backbone (ResNet-50 v1.5) initialized with ImageNet-1K weights. Training uses the Adam optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$), with a base learning rate of 1×10^{-3} and a reduced rate of 1×10^{-6} for activation parameters. Training is capped at 30 epochs with early stopping (patience of 5). The batch size is chosen by a dynamic rule with a small upper bound to prevent unstable updates of the activation parameters. Random seeds and data splits are fixed across runs, and evaluation is performed on a consistent held-out test set using the best validation checkpoint for each method.

A limited hyperparameter search was conducted to determine optimal configurations for the baseline PEFT methods. For Conv-Adapter, the reduction ratio γ was varied, with $\gamma = 16$ yielding the best trade-off between performance and parameter efficiency. For LoRA-C, several rank values were tested, and $r = 4$ was selected as the most effective configuration, balancing expressivity with computational cost.

Evaluation focuses on the macro-averaged F1 score, AUC, and expected calibration error (ECE, L1 binning). This combination captures both predictive performance and confidence alignment; metrics are computed as defined in Section 3.5.

5. Results and Discussion

5.1. Overall Performance and Calibration Trends

Across data regimes, learnable activations provide a better balance of discrimination and calibration than structural PEFT baselines, while using orders of magnitude fewer parameters. In the lowest-data settings (0.5–1%), channel-wise variants are the most reliably calibrated (lower ECE) and remain competitive on F1/AUC while block-shared variants track closely with slightly less variance. Conv-Adapter and LoRA-C show larger swings and tend to be overconfident under sparse supervision, which aligns with their higher effective capacity and sensitivity to hyperparameters.

Macro-F1 Scores (Table 3): F1 increases with data for all methods. A notable pattern is that approaches with very small trainable parameter counts (ReLU-FC-Only and block-shared activations) are surprisingly competitive or best, especially below 10% of the data, suggesting that minimizing the number of degrees of freedom stabilizes learning under scarcity. Channel-wise activations catch up by 10% and above, indicating that fine-grained modulation becomes useful once supervision is sufficient.

Table 3: Macro-averaged F1 scores across PEFT methods and dataset subsets. Performance generally improves with increasing training data, with ReLU FC Only and block-shared activations (SwishL, KGLAP) achieving the strongest overall F1 scores. Conv-Adapter and LoRA-C exhibit consistently lower F1 across subsets, indicating reduced class balance performance under limited data.

Method	0.5%	1%	10%	50%	100%
Relu FC Only	**0.64**	0.75	0.86	0.91	0.93
SwishL (Block-Shared, Stages 3–4)	**0.64**	0.74	0.86	0.92	0.93
KGLAP (Block-Shared, Stages 3–4)	0.51	0.73	0.83	0.86	0.88
KGLAP (Channel-Wise, Stage 4)	0.21	0.73	0.86	0.92	0.93
PReLU (Channel-Wise, Stage 4)	0.28	0.69	0.84	0.91	0.91
Conv-Adapter ($\gamma = 16$)	**0.64**	0.45	0.51	0.87	0.80
Lora-C ($r = 8$)	0.46	0.32	0.33	0.53	0.70

AUC trends (Table 4): AUC measures class separability independent of classification threshold, providing a deeper view of model discrimination than F1, which evaluates performance only at a fixed decision boundary. While F1 reflects how well final class assignments match ground truth, AUC captures how consistently the model ranks true samples above false ones across all possible thresholds, offering a threshold-agnostic perspective on discriminative ability. Here, all methods outperform the simple linear classifier (*ReLU FC Only*), confirming that each PEFT approach improves the model's ability to separate classes even when trained with minimal supervision. Channel-wise activations achieve the highest AUC in the 0.5–1% data regimes, implying that independent channel modulation enhances feature-level separability when data are scarce. Beyond 10% of available training data, differences across methods narrow, as the pretrained backbone dominates the representational capacity and training converges toward similar decision boundaries not explicitly optimized for ranking performance. LoRA-C attains the highest mean AUC overall but exhibits larger variance across subsets, suggesting that while low-rank adaptations can occasionally enhance separability, they are less stable than activation-based approaches under limited data conditions.

Table 4: AUC across PEFT methods and dataset subsets. Discriminative performance generally increases with available training data but saturates beyond 10%. Block-shared activations (KGLAP, PReLU) maintain stable AUC across subsets, while LoRA-C achieves higher variance and Conv-Adapter underperforms overall. These results suggest that activation-based PEFT retains strong class separability even under limited supervision.

Method	0.5%	1%	10%	50%	100%
Lora-C ($r = 8$)	0.45	0.35	0.39	0.42	0.37
KGLAP (Channel-Wise, Stage 4)	0.51	0.36	0.37	0.36	0.35
PReLU (Channel-Wise, Stage 4)	0.50	0.37	0.36	0.35	0.35
KGLAP (Block-Shared, Stages 3–4)	0.48	0.36	0.36	0.36	0.35
Conv-Adapter ($\gamma = 16$)	0.34	0.44	0.40	0.39	0.33
SwishL (Channel-Wise, Stage 4)	0.43	0.34	0.35	0.35	0.35
Relu FC Only	0.42	0.34	0.35	0.35	0.35

Expected Calibration Error (Table 5): ECE quantifies the gap between predicted confidence and actual correctness, with lower values indicating better-calibrated probability estimates. At 0.5–1% data, channel-wise activation-based PEFT methods achieve the lowest calibration error, with SwishL and PReLU performing best, followed closely by ReLU FC Only and block-shared KGLAP. In contrast, LoRA-C and Conv-Adapter exhibit substantially higher ECE, reflecting a tendency toward overconfident predictions despite their larger parameter budgets. As the amount of training data increases, ECE values rise across all methods as models converge but activation-based methods remain comparatively better calibrated. This pattern suggests that learning nonlinearities provides a form of implicit regularization that improves confidence alignment, particularly when supervision is limited.

Table 5: Expected Calibration Error (ECE, lower is better) across PEFT methods and dataset subsets. Activation-based PEFT methods such as channel-wise KGLAP and PReLU achieve substantially lower calibration error than LoRA-C and Conv-Adapter, indicating more reliable confidence estimates. Calibration error tends to increase with data size as models approach convergence, yet activation-based variants consistently preserve superior confidence alignment across all regimes.

Method	0.5%	1%	10%	50%	100%
KGLAP (Block-Shared, Stages 3–4)	0.36	0.43	0.67	0.72	0.80
PReLU (Channel-Wise, Stage 4)	0.30	0.39	0.67	0.84	0.88
SwishL (Channel-Wise, Stage 4)	0.18	0.51	0.79	0.89	0.90
Relu FC Only	0.28	0.51	0.79	0.89	0.90
KGLAP (Channel-Wise, Stage 4)	0.45	0.53	0.78	0.89	0.89
Lora-C ($r = 8$)	0.76	0.79	0.80	0.71	0.79
Conv-Adapter ($\gamma = 16$)	0.84	0.65	0.73	0.95	0.93

Performance summary: Overall, learnable activations deliver a favorable balance of F1, AUC, and ECE with 10^2–$10^3\times$ fewer parameters than Conv-Adapter and LoRA-C. They provide a simple drop-in alternative for low-resource domains where calibration is as important as accuracy.

5.2. Influence of Parameterization and Activation Type

Channel-shared vs. channel-wise parameterization: The distinction between shared and channel-wise activation schemes proves consequential. Channel-wise parameterization improves F1 and ECE, particularly in small subsets, at only a modest parameter cost. Per-channel adaptivity likely provides localized sensitivity control, enabling each feature map to modulate its dynamic range during adaptation.

Activation family effects: A notable finding was the robustness of performance across different activation function designs (PReLU, SwishLearn, KGLAP). While the exact form of the nonlinearity influenced calibration and accuracy slightly, all three learnable variants yielded broadly similar outcomes. Overall, differences among PReLU, SwishLearn, and KGLAP are smaller than those introduced by the injection strategy. Such stability is desirable for practitioners seeking efficient tuning without bespoke activation design, but leaves the door open for future work to explore systematic searches or meta-learning strategies to identify other candidate activation functions.

5.3. Results by Data Regime

Dataset subset = 0.5%: At 0.5% of the training set, channel-wise PReLU and KGLAP achieve the smallest distance to the ideal calibration–discrimination point. These methods maintain both high AUC and low ECE, demonstrating that lightweight activation tuning can extract useful structure from sparse supervision without inducing overconfidence. Conv-Adapter and LoRA-C degrade sharply, confirming that larger architectural modifications are harder to optimize under extreme data scarcity.

Table 6: AUC–ECE tradeoff on 0.5% subset, ranked by Euclidean distance to the ideal point (AUC = 1, ECE = 0) where $d = \sqrt{(1 - \mathrm{AUC})^2 + \mathrm{ECE}^2}$. Smaller d indicates a better joint tradeoff (higher AUC and lower calibration error).

Method	**AUC** ↑	**ECE**$_{L1}$ ↓	**Distance** ↓
PReLU (Channel-Wise, Stage 4)	0.50	0.30	**0.58**
SwishL (Channel-Wise, Stage 4)	0.43	0.18	0.60
KGLAP (Block-Shared, Stages 3–4)	0.48	0.36	0.63
ReLU FC Only	0.42	0.28	0.64
KGLAP (Channel-Wise, Stage 4)	**0.51**	0.45	0.67
Lora-C ($r = 8$)	0.45	0.76	0.94
Conv-Adapter ($\gamma = 16$)	0.34	0.84	1.07

Dataset subset = 100%: At larger subset sizes, differences in AUC narrow considerably across all methods, but activation-based PEFT retains the most stable calibration. This persistence indicates that learnable activations refine confidence estimates even when discrimination has plateaued, highlighting their role as confidence-shaping mechanisms rather than purely feature-learners.

Table 7: AUC–ECE tradeoff on full dataset (100% subset), ranked by Euclidean distance to the ideal point (AUC = 1, ECE = 0) where $d = \sqrt{(1 - \mathrm{AUC})^2 + \mathrm{ECE}^2}$. Smaller d indicates a better joint tradeoff (higher AUC and lower calibration error).

Method	**AUC** ↑	**ECE**$_{L1}$ ↓	**Distance** ↓
Lora-C ($r = 8$)	0.37	0.79	**0.86**
KGLAP (Block-Shared, Stages 3–4)	0.35	0.80	0.87
KGLAP (Channel-Wise, Stage 4)	0.35	0.89	0.92
PReLU (Channel-Wise, Stage 4)	0.35	0.88	0.92
SwishL (Channel-Wise, Stage 4)	0.35	0.90	0.93
ReLU FC Only	0.35	0.90	0.93
Conv-Adapter ($\gamma = 16$)	0.33	0.93	0.99

5.4. Class-Agnostic Tumor Detection

All methods exceed 90% accuracy in high-data regimes. Activation-based PEFT matches the baseline while requiring less than 0.06% of parameters, indicating that calibration improvements do not compromise discriminative power for the clinically relevant binary task. The small advantage of smoother activations (Swish-like) at full data suggests that gradient continuity may further enhance decision reliability near class boundaries.

Table 8: Binary tumor vs. non-tumor classification accuracy across all data regimes. For this metric, all tumor subtypes were merged into a single positive class, such that predictions were considered correct if any tumor type was identified rather than *no tumor*. Most methods exceed 90% accuracy even in low-data settings, demonstrating strong transferability of pretrained features. Activation-based PEFT achieves comparable or superior performance on average, highlighting its efficiency and robustness for clinically relevant tumor detection tasks.

Method	**0.5%**	**1%**	**10%**	**50%**	**100%**
SwishL (Block-Shared, Stages 3–4)	89.96	94.94	96.72	98.15	98.65
SwishL (Channel-Wise, Stage 4)	89.53	94.59	96.72	98.08	98.65
ReLU FC Only	88.60	94.80	96.72	98.08	98.65
KGLAP (Block-Shared, Stages 3–4)	83.26	91.17	94.52	95.51	97.15
KGLAP (Channel-Wise, Stage 4)	77.35	90.46	96.15	98.29	98.58
PReLU (Channel-Wise, Stage 4)	77.92	86.04	94.66	97.44	97.93
Conv-Adapter ($\gamma = 16$)	82.83	84.05	92.24	94.87	86.04
Lora-C ($r = 8$)	78.63	85.26	86.89	85.97	89.60

5.5. Synthesis and Implications

Tuning nonlinearities enables lightweight adaptation of pretrained CNNs to new domains with minimal parameter overhead. Learnable activations improve probabilistic calibration while maintaining competitive discrimination, particularly under limited supervision where structural PEFT methods overfit or fail to converge efficiently. The results indicate that confidence alignment benefits arise primarily from activation adaptivity rather than specific function choice, suggesting that generalized parametric activations can serve as practical, reusable adaptation tools.

5.6. Limitations

This study uses single-seed runs and a single dataset, which limits statistical confidence and external validity. Only classification was evaluated; more complex tasks (e.g., segmentation, multi-label, or multi-modal) may demand additional capacity or earlier-stage tuning. Finally, we explored a small slice of the design space for activation initialization and optimizer coupling; a systematic sweep could further improve stability, convergence, and calibration.

6. Conclusion and Future Work

This paper introduced learnable activation functions as a parameter-efficient fine-tuning strategy for convolutional neural networks. Our experiments on a Brain Tumor MRI dataset demonstrate that activation tuning consistently improves calibration at negligible parameter cost relative to Conv-Adapter and LoRA-C, and preserves competitive discriminatory ability.. These results highlight the potential of learnable activations as a lightweight and effective tool for scenarios where confidence alignment is critical, such as medical imaging.

Several promising directions for future work remain. Evaluation on additional datasets and CNN architectures is required to confirm the generality of our findings, and further exploration of activation configurations is warranted. Moreover, strategies that modify earlier stages of the network or experiment with different parameter initializations can be evaluated with this stable methodology. Finally, despite the resilience of our results to varied activation function forms, the field may benefit from the development of domain-specific activation functions, rather than relying on arbitrary or borrowed designs.

In summary, learnable activations offer a novel balance of efficiency and calibration. By extending these ideas through richer activation designs, future research can build on this foundation toward more effective and practical parameter-efficient transfer learning methods.

Appendix A. Additional Figures

Table A.9: Test performance across training subset sizes. Best per subset and metric in **bold**.

Dataset Subset %	0.5%		1%		10%		50%		100%	
Accuracy, F1 Score	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Relu FC Only	0.40	0.64	0.42	0.75	0.34	0.86	0.92	0.91	0.93	0.93
SwishL (Block-Shared, Stages 3–4)	0.25	0.64	0.42	0.74	0.03	0.86	0.92	0.92	0.93	0.93
SwishL (Channel-Wise, Stage 4)	0.26	0.59	0.43	0.75	0.03	0.87	0.92	0.91	0.93	0.93
KGLAP (Block-Shared, Stages 3–4)	0.19	0.51	0.07	0.73	0.04	0.83	0.03	0.86	0.89	0.88
KGLAP (Channel-Wise, Stage 4)	0.24	0.21	0.09	0.73	0.44	0.86	0.92	0.92	0.93	0.93
PReLU (Channel-Wise, Stage 4)	0.29	0.28	0.39	0.69	0.44	0.84	0.91	0.91	0.92	0.91
PReLU (Block-Shared, Stages 3–4)	0.29	0.26	0.12	0.65	0.05	0.83	0.91	0.90	0.90	0.90
Conv-Adapter ($\gamma = 16$)	0.19	0.64	0.21	0.45	0.24	0.51	0.87	0.87	0.81	0.80
Lora-C ($r = 4$)	0.36	0.33	0.37	0.43	0.14	0.58	0.74	0.71	0.36	0.50
Lora-C ($r = 8$)	0.48	0.46	0.24	0.32	0.21	0.33	0.59	0.53	0.70	0.70

Dataset Subset %	0.5%		1%		10%		50%		100%	
Precision, Recall	Prec	Rec	Prec	Rec	Prec	Rec	Prec	Rec	Prec	Rec
Relu FC Only	0.66	0.64	0.76	0.76	0.86	0.86	0.92	0.92	0.93	0.93
SwishL (Block-Shared, Stages 3–4)	0.66	0.63	0.75	0.75	0.87	0.86	0.92	0.91	0.93	0.93
SwishL (Channel-Wise, Stage 4)	0.62	0.58	0.76	0.76	0.87	0.87	0.92	0.91	0.93	0.93
KGLAP (Block-Shared, Stages 3–4)	0.61	0.52	0.75	0.74	0.83	0.83	0.86	0.86	0.89	0.88
PReLU (Channel-Wise, Stage 4)	0.52	0.33	0.71	0.69	0.84	0.84	0.91	0.90	0.92	0.91
KGLAP (Channel-Wise, Stage 4)	0.42	0.31	0.74	0.73	0.86	0.86	0.92	0.92	0.93	0.93
PReLU (Block-Shared, Stages 3–4)	0.42	0.33	0.67	0.65	0.83	0.91	0.91	0.90	0.90	0.90
Conv-Adapter ($\gamma = 16$)	0.65	0.64	0.47	0.52	0.56	0.88	0.88	0.87	0.85	0.80
Lora-C ($r = 4$)	0.50	0.42	0.62	0.49	0.60	0.81	0.81	0.73	0.48	0.57
Lora-C ($r = 8$)	0.61	0.48	0.51	0.45	0.45	0.60	0.62	0.60	0.74	0.70

Table A.10: AUC and ECE across training subset sizes. Lower ECE is better. Best per subset and metric in **bold**.

Dataset Subset %	0.5%		1%		10%		50%		100%	
AUC, ECE	AUC	ECE	AUC	ECE	AUC	ECE	AUC	ECE	AUC	ECE
Lora-C ($r = 4$)	0.52	0.61	0.44	0.81	0.34	0.86	0.33	0.91	0.37	0.81
Lora-C ($r = 8$)	0.45	0.76	0.35	0.79	0.39	0.80	0.42	0.71	0.37	0.79
KGLAP (Channel-Wise, Stage 4)	0.51	0.45	0.36	0.53	0.37	0.78	0.36	0.89	0.35	0.89
PReLU (Channel-Wise, Stage 4)	0.50	0.30	0.37	0.39	0.36	0.67	0.35	0.84	0.35	0.88
KGLAP (Block-Shared, Stages 3–4)	0.48	0.36	0.36	0.43	0.36	0.67	0.36	0.72	0.35	0.80
PReLU (Block-Shared, Stages 3–4)	0.46	0.46	0.33	0.44	0.38	0.69	0.37	0.85	0.37	0.83
Conv-Adapter ($\gamma = 16$)	0.34	0.84	0.44	0.65	0.40	0.73	0.39	0.95	0.33	0.93
SwishL (Channel-Wise, Stage 4)	0.43	0.18	0.34	0.51	0.35	0.79	0.35	0.89	0.35	0.90
SwishL (Block-Shared, Stages 3–4)	0.43	0.27	0.34	0.51	0.35	0.79	0.35	0.89	0.34	0.90
Relu FC Only	0.42	0.28	0.34	0.51	0.35	0.79	0.35	0.89	0.35	0.90

Table A.11: Class-agnostic tumor accuracy (%) by method and training subset. For this metric, models were scored equally regardless of what type of tumor was detected, so long as a tumor was present.

Method	**0.5%**	**1%**	**10%**	**50%**	**100%**
SwishL (Block-Shared, Stages 3–4)	89.96	94.94	96.72	98.15	98.65
SwishL (Channel-Wise, Stage 4)	89.53	94.59	96.72	98.08	98.65
ReLU FC Only	88.60	94.80	96.72	98.08	98.65
KGLAP (Block-Shared, Stages 3–4)	83.26	91.17	94.52	95.51	97.15
KGLAP (Channel-Wise, Stage 4)	77.35	90.46	96.15	98.29	98.58
PReLU (Channel-Wise, Stage 4)	77.92	86.04	94.66	97.44	97.93
PReLU (Block-Shared, Stages 3–4)	74.72	80.91	93.59	97.36	96.94
Conv-Adapter ($\gamma = 16$)	82.83	84.05	92.24	94.87	86.04
Lora-C ($r = 8$)	78.63	85.26	86.89	85.97	89.60
Lora-C ($r = 4$)	75.71	81.55	72.58	93.52	72.51

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